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A Reproducible Framework and a Preregistered Temporal Validation

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Reconstructing Journal Thematic Identity from Article-Level Topics: A Reproducible Framework and a Preregistered Temporal Validation

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Abstract

Journals are conventionally described by editorial subject classifications, yet what a journal actually publishes need not coincide with its assigned label. We present a framework for reconstructing a journal's thematic identity from article-level evidence while maintaining a strict separation between what is observed and what is inferred. From the distribution of OpenAlex subfields across a journal's articles we define an observed thematic representation, and from it three comparative descriptors—alignment with the editorial classification, intrinsic coverage of the thematic nucleus, and relative position—that yield a two-dimensional typology. Over a transparently constructed cohort of journals we characterize thematic concentration, temporal stability, and the discriminant structure of the descriptors. We then submit the framework to a preregistered prospective replication that shifts the observation window forward in time, and to a documentary control that recomputes the same analysis on document- rather than citation-weighted distributions. The temporal replication weakens, but does not overturn, the expected stability pattern; the documentary control shows that the large-scale reorganization observed under citation weighting is substantially attenuated when document weighting is used, while leaving a residual whose interpretation we deliberately leave open. All thresholds were fixed in advance, and code, data, preregistrations, and a code-to-documentation audit are released with cryptographic version identifiers. The result is a reproducible, transparent method that profiles journals thematically without mistaking an inferred description for an intrinsic property.

1. Introduction

Can a journal's thematic identity be reconstructed from the topics of its published articles rather than from its editorial classification? Editorial subject classifications—Scopus All Science Journal Classification (ASJC) categories, Web of Science subject areas, and similar schemes—assign each journal to one or more predefined fields and are indispensable for indexing, evaluation, and discovery. But they are editorial artifacts: assigned at the level of the journal, revised infrequently, and coarse relative to the thematic granularity at which research is actually organized. What a journal publishes, article by article, need not coincide with the label it carries. A journal nominally placed in one subfield may draw the bulk of its content, or of its citation impact, from another; its thematic center of gravity may drift over time; and two journals sharing an editorial category may cover markedly different ground. Three facts, taken together, turn this from a convenience into a requirement: editorial classifications are assigned to the journal as a whole and revised infrequently; the articles a journal publishes carry their own, finer-grained topics; and the distribution of those topics shifts as a journal's content evolves. It follows that a journal's thematic identity can be neither assumed constant nor deduced from its editorial label alone—it must be reconstructed from what the journal actually publishes, and reconstructed again as that content changes. Reconstructing a journal's thematic profile from article-level evidence is therefore not one option among several but a natural and long-standing aim of scientometrics.

The availability of large open bibliographic corpora makes such reconstruction feasible at scale. OpenAlex assigns each work a primary topic within a hierarchical taxonomy (topic → subfield → field → domain), so that the thematic content of a journal can be summarized as a distribution over subfields aggregated across its articles (Priem et al., 2022). This distribution is an attractive basis for profiling, but it also invites a category error that we take pains to avoid. It is tempting to read the resulting profile as the journal's underlying thematic identity, or to treat a mismatch between profile and editorial category as evidence that the classification is wrong. Both moves overreach. The distribution is an observation; a journal's "identity" is an inference drawn from it, conditioned on the taxonomy, the aggregation rule, the observation window, and the data snapshot.

The framework we present is organized around this distinction. We separate the observed thematic representation of a journal—the distribution itself, together with descriptors computed directly from it—from any inferred thematic identity, and we adopt as a governing principle that every inference about thematic identity must be constructed from an observed representation and must never substitute for it (Principle 1). Under this principle, a journal's identity is reported explicitly as an inference, never asserted as an intrinsic property and never presented as a correction of the editorial scheme; the editorial classification is treated as a reference for comparison rather than an object to be corrected. This is not merely a rhetorical safeguard. It determines what the descriptors are permitted to claim, where the boundary between deductive and empirical results falls, and how the method may be embedded in downstream systems—research evaluation, longitudinal tracking of thematic change, dynamic classification, information retrieval, and current research information systems—without silently reinterpreting the science.

Our contributions are four. First, we formalize the framework: an observed thematic representation obtained by applying an aggregation rule to a journal's article-level topic assignments, and three comparative descriptors—alignment with the editorial classification, intrinsic coverage of the thematic nucleus, and relative position—together with propositions that make explicit which of their properties are deductive consequences of the framework and which are empirical claims requiring evidence. Second, we characterize, over a transparently constructed cohort of journals, the thematic concentration, temporal stability, and discriminant structure of these descriptors, and we summarize the comparable cohort in a two-dimensional typology. Third, we submit the framework to a preregistered prospective replication that shifts the observation window forward in time, complemented by a documentary control that recomputes the analysis on document-weighted rather than citation-weighted distributions; the control is designed to separate genuine thematic reorganization from the maturation of recent citations. Fourth, we release the method as a reproducible protocol: thresholds fixed in advance, code and data deposited under persistent identifiers, and a code-to-documentation audit with cryptographic version identifiers linking every reported number to the implementation that produced it.

We regard the last contribution as inseparable from the others. A framework whose central claim is a separation between observation and inference is only as trustworthy as the discipline with which it is applied; preregistration and reproducibility are the mechanisms that keep the separation honest when results are unfavorable. Indeed, the temporal replication reported below returns an inconclusive verdict under its prespecified decision rule, and we report it as such, without reinterpretation—an outcome the reproducibility apparatus was built to make possible rather than to avoid. The remainder of the paper introduces related work (§2), the conceptual framework (§3), the method and metrics (§4), the experimental design and cohorts (§5), results (§6), and discussion, limitations, and conclusions (§7–§9).

2. Related work

Journal-level classification and its limits. Most bibliometric infrastructure organizes journals through editorial classification schemes—the Scopus ASJC (Baas et al., 2020), the Web of Science subject categories, and comparable national or disciplinary taxonomies. These schemes are assigned at the journal level, revised infrequently, and deliberately coarse; the mismatches they create between a journal's category and its content, and the difficulties they pose for field delineation and indicator normalization, are well documented (Waltman & van Eck, 2012; Milojević, 2020). These critiques motivate descriptions of journals grounded in what they publish rather than in how they are labeled.

Content- and citation-based profiling. A substantial literature recovers journal and field structure from data: co-citation analysis (Small, 1973), bibliographic coupling and direct citation, and global and local maps of science (Boyack & Klavans, 2010; van Eck & Waltman, 2010), including journal maps derived from citation relations (Leydesdorff, 2006). OpenAlex has broadened access to content-based profiling by attaching to every work a primary topic within a hierarchical taxonomy derived from large-scale classification of metadata and citations (Priem et al., 2022); work in the science-of-science tradition increasingly builds on OpenAlex- and SciSciNet-scale corpora to examine field structure, classification, and citation behavior (e.g., Milojević, 2020; Algaba et al., 2026). These approaches characterize thematic structure with considerable success, but they generally stop at descriptive profiling: the profile is reported as a picture of the journal, not examined as an inference conditioned on how it was produced.

Concentration, diversity, and stability. Orthogonally to classification, scientometrics has long measured how concentrated or diverse a unit's output is, from concentration indices such as the Herfindahl–Hirschman index and Shannon entropy to distance-weighted diversity measures that jointly capture variety, balance, and disparity (Stirling, 2007; Porter & Rafols, 2009; Leydesdorff & Rafols, 2011). These indicators describe the shape of a distribution but say nothing, on their own, about whether that distribution should be read as an identity or about how stable it is over time.

Reproducibility and preregistration. Finally, a growing methodological literature urges open data and code and, more recently, preregistration and analytic transparency as responses to the flexibility inherent in bibliometric analysis; this journal's founding under Fair Open Access principles is itself part of that movement (Nosek et al., 2018). Our transparency protocol is designed in that spirit.

What is missing. Our framework builds on these lines of work while posing a different methodological question. Existing work reconstructs thematic structure, improves classification, or measures interdisciplinarity; none of it explicitly separates the observed thematic representation from the inferred thematic identity as distinct epistemic objects subjected to preregistered validation. Across these strands, journals are either assigned a classification or profiled descriptively, and concentration or diversity is reported as a summary statistic. What has not been done, to our knowledge, is to treat a journal's thematic identity as an explicitly *inferred* object—derived from the article-level topic distribution, conditioned on taxonomy, aggregation rule, window, and snapshot, and held formally distinct from the observed representation on which it rests—nor to subject such an inference to a preregistered temporal validation with released reproducibility artifacts. This paper addresses that gap. Concretely, it asks: **can a journal's thematic identity be inferred from the thematic distribution of its articles, and how does that inference behave under a forward shift of the observation window and under a change in how article mass is weighted?**

3. Conceptual framework

We distinguish a methodological framework from the representations it produces. The framework (Definition 1) comprises a topic taxonomy, an aggregation rule, and the conditions under which they may be applied to a journal's articles. Although we instantiate it with the OpenAlex taxonomy throughout, the framework is defined over any hierarchical topic taxonomy and does not depend on that particular choice; how sensitive the resulting descriptors are to the taxonomy is an empirical question in its own right, examined in part through the taxonomic-perturbation axis (§5–§6). Its output for a journal *j* is a thematic representation (Definition 2): R_j = A(D_j), where D_j is the collection of article-level topic assignments of the journal within an observation window and A is an aggregation rule returning a distribution over subfields. The rule admits two mass modes that we keep strictly separate throughout: a document-weighted mode, in which each subfield's mass is the share of the journal's documents assigned to it (p_doc), and a citation-weighted mode, in which mass is the corresponding share of citations (p_cit). R_j is an observation; it is not, and is not read as, the journal's identity. In the vocabulary of measurement, R_j is a measurement of a journal's thematic content rather than that content itself—a representation is not the object it represents—and reading the measurement as the journal's identity is precisely the category error the framework is built to prevent.

This separation is elevated to a governing principle, which we call the **observation–inference separation principle** (Principle 1): a representation is observed; an identity is inferred. In full, observed representations constitute evidence, and scientific identities are inferences derived from that evidence. Applied to the object at hand, every inference about a journal's thematic identity must be constructed from an observed representation and must never substitute for it. In consequence, identity is exposed only as an inference explicitly conditioned on the taxonomy, aggregation rule, window, and snapshot; it is never asserted as an intrinsic property of the journal, and a divergence between R_j and the editorial classification is treated as information for comparison, not as a correction of that classification (Figure 1).

From R_j we define the thematic nucleus as the set of subfields accumulated, in order of decreasing mass, until the cumulative mass first reaches a threshold $\tau = 0.5$; the subfield whose inclusion reaches the threshold belongs to the nucleus. Three comparative descriptors follow. Alignment (JCA) is the mass of R_j that falls within the journal's editorial categories E_j. Coverage (M) is the extent to which the nucleus is intrinsically accounted for. Position (D) locates the journal as nuclear, peripheral, or displaced relative to its editorial assignment. Coverage and position together define a two-dimensional typology of the comparable cohort.

We state two kinds of proposition and keep their status distinct; formal statements and proofs are given in Supplementary Material §S1. The **deductive** propositions establish that, given the framework, the observed representation and the three descriptors exist and are derivable from R_j alone—without recourse to the editorial classification for the representation itself. The **empirical** propositions—that coverage and position are non-redundant, and that alignment is not reducible to a journal's structural properties—are claims about the data, not consequences of the definitions, and are evaluated in §6. Maintaining this boundary is what allows the framework to make modest, checkable claims: the existence of a reproducible description is deductive, whereas any statement about what that description reveals is empirical and is held to the corresponding standard of evidence.

4. Method

This section defines the observed representation and the quantities computed from it. Exact formulas and the reference implementation are given in Supplementary Material §S2; the definitions here are those needed to read the results.

Thematic representation. For a journal *j* and an observation window, we take the OpenAlex subfield of the primary topic of each of the journal's works and aggregate these assignments into a distribution over subfields, R_j. We compute R_j under two mass modes, kept separate throughout: in the document-weighted mode (p_doc) a subfield's mass is the fraction of the journal's works assigned to it; in the citation-weighted mode (p_cit) it is the fraction of citations received by works in that subfield. The two modes answer different questions—what a journal publishes versus what of it is cited—and are never combined.

Nucleus and concentration. Ordering subfields by decreasing mass, the thematic nucleus is the shortest leading set whose cumulative mass reaches $\tau = 0.5$, including the subfield at which the threshold is crossed. We summarize the shape of R_j with standard statistics: a concentration index (the sum of squared masses, equivalent to the Herfindahl–Hirschman index) and Shannon entropy, in raw and size-normalized form (exact forms, including logarithm bases, in §S2).

Comparative descriptors. Three descriptors compare R_j either to itself or to the journal's editorial categories E_j. Alignment (JCA) is the citation-weighted mass of R_j falling within E_j. Coverage (M) is the share of the nucleus that is intrinsically accounted for. Position (D) classifies the journal, relative to E_j, as nuclear, peripheral, or displaced. Alignment, coverage, and position are the quantities entering the typology and the discriminant analysis of §6.

Impact. Where a journal-level impact value is required we use a subfield-normalized citation score (JSS) obtained from the same corpus; it is read directly from the pipeline and is not re-derived here.

Temporal stability. To quantify how much a journal's representation moves from one year to the next, we compute, for each pair of consecutive years, the Jensen–Shannon divergence (JSD) between the years' document-weighted distributions, and define stability S as one minus the mean year-over-year divergence. We use the document-weighted distribution for stability specifically to avoid the citation-lag bias that would otherwise contaminate the most recent years.

Comparing two windows. The perturbation analyses of §5 compare a journal's representation between two observation windows. For each journal we report the Jensen–Shannon divergence between its distributions in the two windows (a continuous, whole-distribution measure of movement), the Jaccard similarity of its two nuclei (a set-membership measure), whether its dominant subfield is preserved, and the cross-journal correlations of the concentration and entropy statistics between windows. These four quantities are computed identically in both mass modes; the only difference between a citation-weighted and a document-weighted analysis is the column of mass that enters the distribution.

Implementation. All quantities are produced by a single canonical implementation, which operationalizes the definitions presented here and serves as the executable reference for the framework: every reported quantity is generated by that implementation, and no library substitute is treated as equivalent without an explicit parity check. The Supplementary Material documents each definition against the reference implementation and records version identifiers for the code and for every output.

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5. Experimental design

Cohort. We construct the cohort from OpenAlex sources in a provenance-first manner. A sampling frame of journals meeting fixed eligibility criteria—a calculable subfield-normalized impact (JSS), an ISSN present in the Scopus title list, and sufficient content—yields 1,461 journals across six subfields. Of these, 521 additionally satisfy the conditions for editorial comparison and form the comparable cohort, while the full analytic cohort of

522 adds one additional journal retained exclusively as an illustrative example, excluded from all aggregate estimates and from the descriptor matrix (Figure 2). Analyses that require the editorial classification (alignment, coverage, position, and the typology) are reported on the 521-journal comparable cohort; analyses of the intrinsic representation (concentration, stability, and the perturbation tests) use the analytic cohort. Reported cohort sizes are properties of each run and are recorded in its manifest; no cohort size is hard-coded in the computation.

Perturbation axes. We probe the framework along distinct axes, each isolating a different source of variation. A backward temporal shift and a taxonomic coarsening test whether the descriptor structure is reproduced under changes of window and of taxonomic granularity. A one-day re-extraction—a snapshot with near-zero perturbation—tests the operational reproducibility of the pipeline rather than any substantive hypothesis, and is reported as such. The forward temporal shift, which is the focus of §6.4, moves the window one year ahead so that it incorporates the most recent year of data; it is a replication of the backward test on the same temporal axis, not an independent axis, and its window overlaps the original by roughly three years.

Preregistration. Before executing the forward replication, we fixed—in a versioned protocol—the metrics, the reference values inherited from the backward test, the decision thresholds, and the four-cell rule that maps the outcome onto a verdict; a perturbation-magnitude diagnostic was added, also in advance, to establish that the replication actually displaced content rather than reproducing a near-identical window. We state plainly what this buys: **no exploratory threshold tuning was performed after observing the data**, and the verdict follows mechanically from the frozen rule. The protocol further specifies that a verdict is registered whatever the outcome, so that the replication is a genuinely prespecified test and not a validation retained only when favorable.

Documentary control. Because the representation can be weighted by documents or by citations, and because the forward window necessarily includes the least mature citation year, we complement the replication with a documentary control: the same comparison, on the same windows and the same journals, recomputed with document- rather than citation-weighted mass. The control is designed to separate genuine thematic reorganization from citation maturation. Because the decision grid was calibrated on citation-weighted reference values, it is not transferred as a verdict to the document-weighted analysis; instead the documentary outcome is classified, under a separately preregistered rule, by comparison to two empirical anchors—the stable backward test and the reorganized forward test—with an intermediate inconclusive region that is reported as such rather than forced into either category. The primary comparison is made on the panel of journals eligible in both mass modes, so that any difference reflects the change of weighting and not a change in sample composition.

Reproducibility protocol. The analysis is released as a reproducible unit. Preregistration documents, the reference implementation, output manifests, and a code-to-documentation audit—each carrying a SHA-256 version identifier—are deposited with the paper; the Supplementary Material describes the protocol and the audit, allowing every reported result to be traced back to the exact implementation and execution that produced it.

6. Results

We report results in the order in which they build the argument: the intrinsic representation is stable (§6.1) and its descriptors carry non-redundant structure (§6.2); this structure is reproduced under changes of window and taxonomy (§6.3); and it is then subjected to the preregistered forward replication and its documentary control (§6.4), which is where the paper's central result lies.

6.1 Temporal stability

For each journal we measured how much its document-weighted representation changes from one year to the next and summarized this as a stability value S (§4). Across the analytic cohort the median stability is 0.783 (N = 511), indicating that a journal's year-to-year thematic representation moves little: the bulk of the distribution is preserved from one year to the next, and instability is confined to a minority of journals (Figure 4). Because stability is computed on document- rather than citation-weighted mass, this figure is not inflated by the slow accrual of citations to recent work. Stability therefore describes a genuinely persistent object, which is the precondition for asking whether that object also reproduces under the more demanding perturbations that follow.

6.2 Typology and discriminant structure

The comparable cohort (N = 521) supports the two empirical propositions of §3. First, coverage (M) and position (D) are non-redundant: they place journals in a two-dimensional typology whose cells are populated rather than collapsing onto a single diagonal, so that coverage does not, on its own, fix where the journal sits relative to its editorial assignment. Second, alignment (JCA) is not reducible to a journal's structural properties: regressing alignment on its editorial breadth (the number of assigned categories), its size (the logarithm of its number of works), and its intrinsic shape yielded $R^2 = 0.395$ for the entropy specification and $R^2 = 0.496$ for the concentration specification, leaving a substantial share of cross-journal variance unexplained—so alignment with the editorial classification carries information those structural descriptors do not (Figure 3). The cohort is thematically concentrated—the leading subfield carries close to half of a typical journal's mass—yet this concentration does not by itself fix the comparative descriptors, which is precisely why the typology is informative. These findings establish the empirical propositions as claims the data support, in contrast to the deductive propositions, which hold by construction.

6.3 Robustness to window and taxonomy

Two perturbations test whether the descriptor structure is an artifact of a particular window or taxonomic resolution. A backward temporal shift, which slides the observation window one year into the past, and a taxonomic coarsening, which aggregates subfields to a broader level, both reproduce the descriptor structure: the typology and the separation among descriptors survive these changes. A third check, a one-day re-extraction of the same window, was included as a test of operational reproducibility rather than of any hypothesis; as expected, it produced a near-zero perturbation (year-over-year divergence indistinguishable from zero), confirming that the pipeline returns identical results on identical inputs. We report this last check for completeness and draw no substantive inference from it.

6.4 Preregistered forward replication and documentary control

The central test moves the observation window one year forward, so that it incorporates the most recent year of data, and asks whether the pattern seen in the backward replication reproduces. Because this is the analysis for which every threshold and the verdict rule were fixed in advance, we report its outcome exactly as the frozen rule returns it.

The perturbation was real. Before interpreting any verdict, the preregistered magnitude diagnostic confirms that the forward window genuinely displaced content: every journal in the common panel (N = 506; two entered and eight left the eligibility set between windows) shows a non-zero divergence, and only 2.6% show a divergence below 0.01. The replication is therefore a discriminating test, not a near-identical re-run of the original window.

Under citation weighting, the expected asymmetry weakens. On the citation-weighted representation, the structural criterion is not met: the median mass divergence rises to 0.071 (against 0.046 in the backward replication), and the nucleus is preserved with Jaccard ≥ 0.90 in only 52.6% of journals (against essentially all in the backward replication), even though the dominant subfield persists in 85.0% of cases. Meanwhile the shape descriptors remain highly correlated across the two windows (concentration r = 0.852; raw and normalized entropy r = 0.896 and 0.827). The frozen four-cell rule maps this combination—structure not reproduced, form not at ceiling—onto a verdict of *weakens*: the asymmetry by which structure was more stable than form in the backward replication does not reproduce in the forward window. We record this verdict without reinterpretation.

Under document weighting, the large-scale movement is substantially attenuated. Recomputing the identical comparison with document- rather than citation-weighted mass, on the panel of journals eligible in both modes, changes the picture markedly (Table 1 and Figure 5). The median mass divergence falls to 0.028—below even the stable backward-replication anchor of 0.046—the shape correlations rise to 0.966–0.968, and the dominant subfield persists in 95.8% of journals. The nucleus improves as well, to Jaccard ≥ 0.90 in 72.9% of journals, but does not return to the near-complete preservation of the stable pattern. Under the separately preregistered, anchor-based rule for the documentary control—which compares the document-weighted result to the stable and reorganized citation-weighted anchors and reports an intermediate outcome as inconclusive rather than forcing it into either category—this result falls in the intermediate inconclusive region. The mass divergence points unambiguously to documentary stability; the nucleus does not fully return to it; the two indicators disagree; and the preregistered rule therefore returns an inconclusive classification rather than a verdict of confirmation or refutation.

Table 1. Forward replication on the common panel (N = 506), citation- versus document-weighted mass.

Quantity	Citation-weighted	Document-weighted
Median mass divergence (JSD)	0.071	0.028
Nucleus preserved (Jaccard ≥ 0.90)	52.6%	72.9%
Dominant subfield persists	85.0%	95.8%
Shape correlation (concentration / entropy / norm.)	0.852 / 0.896 / 0.827	0.968 / 0.968 / 0.966
Journals with divergence ≤ 0.01	2.6%	10.1%

Read together, the two panels of Table 1 are the paper's principal empirical finding, and their interpretation is the subject of §7.

7. Discussion

The forward replication produces a result that a less disciplined design would have been free to present in either of two convenient ways—as a confirmation of thematic reorganization or as its absence. The preregistered rule permits neither. What the evidence establishes, and what it leaves open, are different things, and the value of the design is that it keeps them apart.

What the evidence establishes. The large-scale reorganization visible under citation weighting is substantially attenuated under document weighting. The mass divergence, the shape descriptors, and the dominant subfield all return to, or exceed, the stable pattern when the representation is weighted by documents rather than citations. This is consistent with a simple mechanism: the forward window necessarily contains the most recent year of data, whose citations are the least mature, and citation weighting is sensitive to that immaturity in a way that document weighting is not. On this reading, much of what looked like thematic movement under citation weighting reflects the accrual of citations rather than a change in what journals publish. This is a substantive result, and it is one the documentary control was specifically designed to obtain.

What the evidence leaves open. One quantity does not return to the stable pattern: the composition of the nucleus, measured by the Jaccard similarity of the two nuclei, remains short of near-complete preservation even under document weighting. This residual is the one place where the forward window may reflect genuine documentary change in what journals cover. But on the current evidence it cannot be distinguished from an alternative that has nothing to do with the bibliometric phenomenon, and here a property of the observables themselves must be made explicit.

Continuous versus threshold-based observables. The mass divergence and the nucleus Jaccard measure different things and respond to different kinds of change. The divergence is a continuous functional of the whole distribution: it moves in proportion to how much probability mass is redistributed. The nucleus, by contrast, is defined by a hard threshold—the set of subfields accumulated until cumulative mass first reaches one half—and its Jaccard is a set-membership statistic. A subfield sitting near that threshold can enter or leave the nucleus under a change in mass far too small to move the divergence appreciably. A threshold-based observable is therefore intrinsically more volatile than a continuous one, and a stable mass distribution can coexist with partial turnover in nucleus membership. The residual we observe is exactly the signature that this discreteness would produce, and it is also exactly the signature that a genuine, small documentary reorganization concentrated near the nucleus boundary would produce. The two are not separable with the data at hand. We stress that this decoupling is a property of the operators, not evidence of an inconsistency in the measurement; presenting it as such is what prevents the residual from being read either as a confirmation or as a contradiction.

What is actually interesting here. The result of scientific interest is not that the method "works." It is that a design fixed entirely in advance was able to separate a claim the evidence supports—that the apparent reorganization is largely an artifact of citation maturation—from a claim it does not yet warrant—that the nucleus residual reflects real documentary change rather than the discreteness of a threshold operator. Throughout, the journal's identity is reported explicitly as an inference: a description conditioned on the taxonomy, the weighting, and the window, held apart from the observed representation on which it rests. The distinction is fundamental: a representation is observed; an identity is inferred. The framework earns its keep not when the replication confirms a hypothesis but here, when it does not, and the separation between observation and inference survives the inconclusiveness intact. Its contribution, in the end, is not to redefine what journals are but to discipline how thematic evidence about them is to be read.

8. Limitations

The most consequential limitation is the one the previous section names: the nucleus residual under document weighting is genuinely ambiguous between a real, small documentary reorganization and the discreteness of a threshold operator. Resolving it requires a further test designed for that purpose—a null model that asks how much nucleus turnover a mass distribution as stable as the one observed would produce by threshold discreteness alone, together with a directionality analysis that asks whether the subfields entering and leaving nuclei are idiosyncratic across journals or systematic. The preregistered design intentionally prevents resolving this ambiguity post hoc; the discriminating experiment is specified but deferred, and we report the present result as inconclusive rather than settle it after the fact.

Several further limitations bound the scope of the claims. The cohort spans six subfields and a single comparable frame, so the typology and stability figures describe this cohort and are not offered as population estimates. All quantities inherit the taxonomy, coverage, and snapshot of the underlying corpus; a different taxonomy or a later snapshot could shift subfield boundaries and, through them, the nucleus. The forward replication is a replication on the temporal axis rather than an independent axis of variation, and the citation-lag sensitivity we exploit as a control is also, in other analyses, a source of bias to be managed. Finally, the descriptors compare a journal to its editorial classification and are only defined for journals for which that classification is available; content-only journals are described but not compared, and are excluded from the typology accordingly. None of these limitations affects the central methodological claim, but each conditions the empirical statements the framework is entitled to make.

9. Conclusions

We set out to ask whether a journal's thematic identity can be reconstructed from the topics of its articles rather than from its editorial classification, and how that reconstruction behaves under perturbation. The answer is that it can be reconstructed as an observed representation and characterized by descriptors that are stable, non-redundant, and reproducible under changes of window and taxonomy—and that its behavior under a forward temporal shift is informative precisely because the design refused to overstate it. The reorganization apparent under citation weighting is largely attenuated under document weighting; a residual in nucleus composition remains, and its interpretation is left open pending a test built to decide it.

The principal contribution of this work is not a new indicator but a methodological discipline for separating observed thematic representations from inferred thematic identity—a framework that requires the two to be kept apart, and that holds to that separation even when the results are inconclusive. The descriptors, the typology, and the stability characterization are useful, but they are instruments of a discipline whose value is clearest when the evidence is ambiguous: a journal's identity is reported explicitly as an inference, never as an intrinsic property, and never as a correction of the classification against which it is compared. The method underlies an analytics application, but its scientific contribution stands independently of that use—as a reproducible way of describing what journals are about that is honest about the boundary between what such a description observes and what it infers. Although developed and illustrated for journals, the separation between observed representation and inferred identity is not specific to them; it applies in principle to any bibliometric entity described through aggregated evidence—researchers, institutions, fields—though we neither pursue nor claim those extensions here. Scientific identities are not observations; they are inferences drawn from observed evidence under explicitly stated conditions.

A representation is observed; an identity is inferred.

Figures

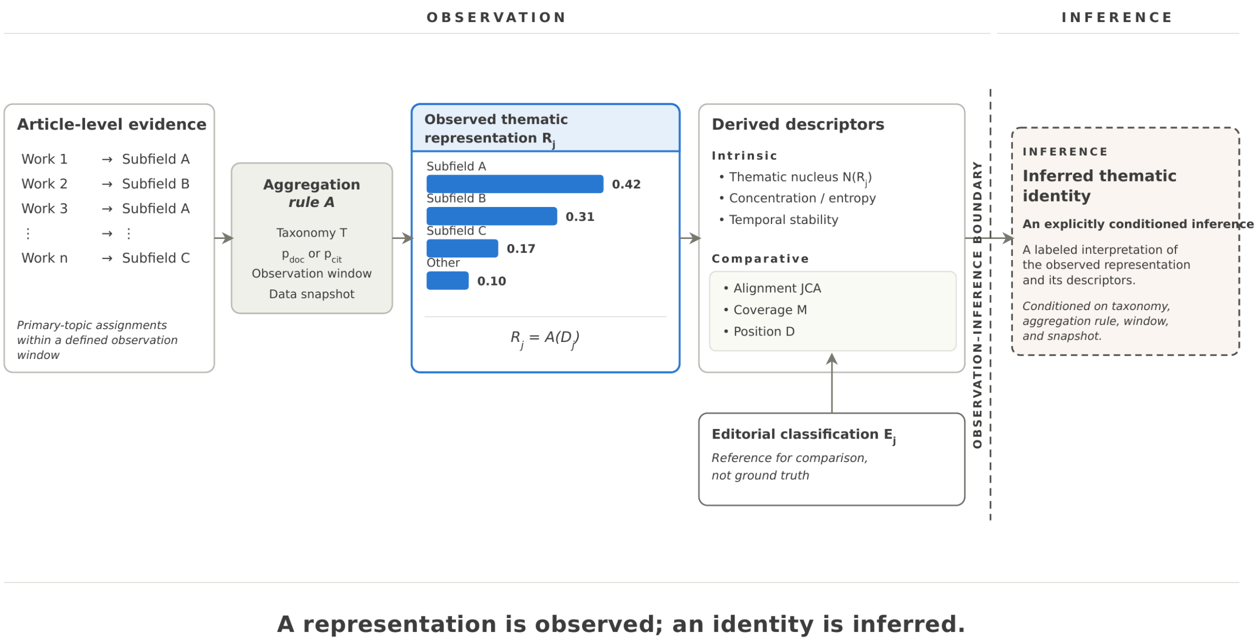

Figure 1 (produced; vector PDF available). *From article-level evidence to inferred journal thematic identity.* Article-level topic assignments are aggregated under an explicitly specified taxonomy, aggregation rule, observation window, data snapshot, and mass mode to produce the observed thematic representation R_j. Intrinsic and comparative descriptors are derived from this representation; the editorial classification E_j enters only as a reference for comparison and does not determine R_j. The journal's thematic identity is subsequently reported as an inference conditioned on the observed representation and its construction conditions, rather than as an intrinsic property or a correction of the editorial classification. The dashed boundary marks the transition from observation to inference: a representation is observed; an identity is inferred.

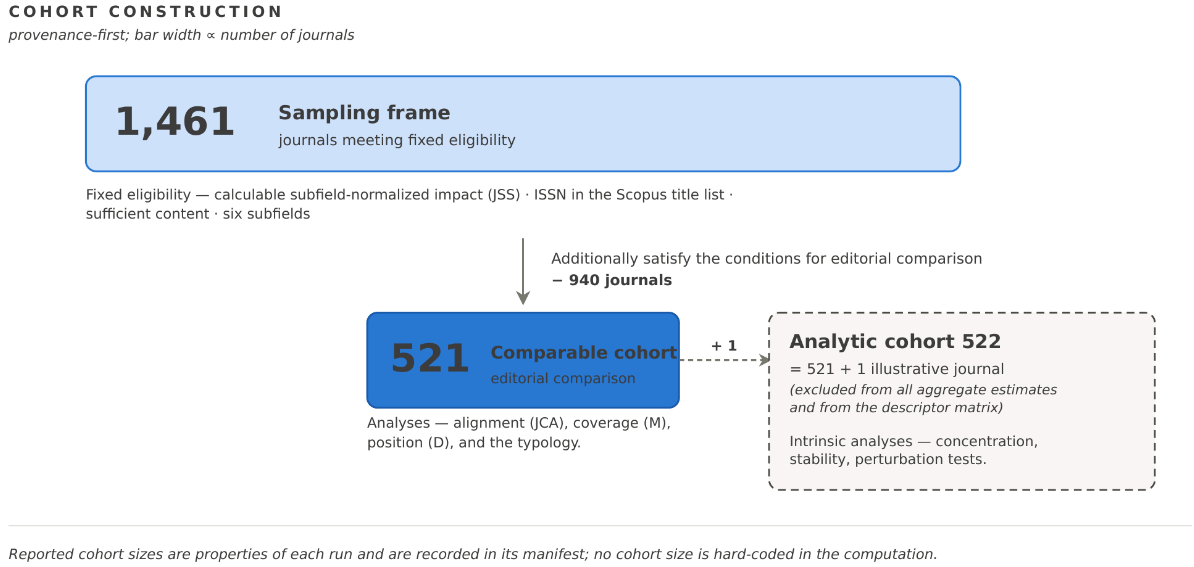

Figure 2 (produced; vector PDF available). *Cohort construction.* A sampling frame of 1,461 journals meeting fixed eligibility (calculable subfield-normalized impact, ISSN in the Scopus title list, sufficient content; six subfields) reduces to the 521-journal comparable cohort that additionally satisfies the conditions for editorial comparison, used for the editorial-comparison analyses (alignment, coverage, position, and the typology). The analytic cohort of 522 adds one illustrative journal—excluded from all aggregate estimates and the descriptor matrix—and supports the intrinsic analyses (concentration, stability, perturbation). Bar width is proportional to the number of journals; reported sizes are recorded in each run's manifest and are not hard-coded.

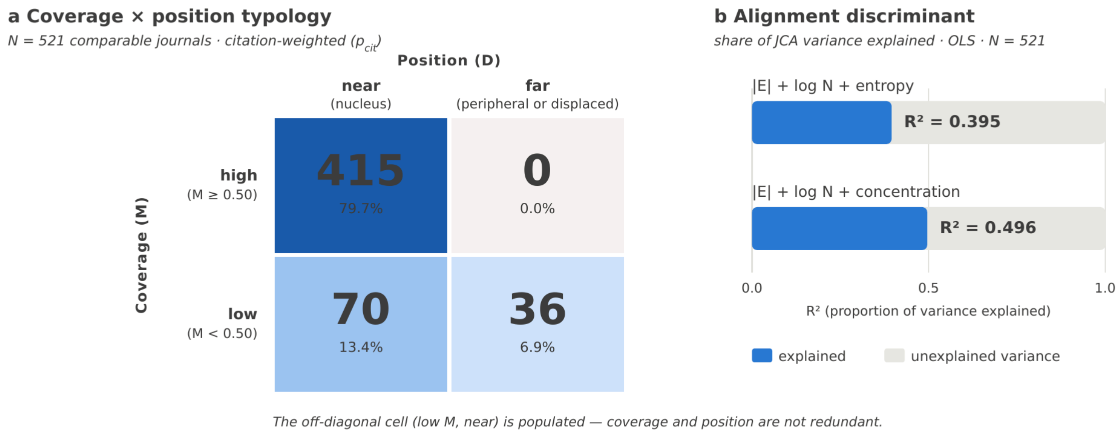

Figure 3 (produced; vector PDF available). *Coverage × position typology and the alignment discriminant.* **(a)** The comparable cohort (N = 521) placed in a 2×2 of coverage M (high, M ≥ 0.50; low, M < 0.50) against position D (near, editorial set within the nucleus; far, peripheral or displaced); counts 415 / 0 / 70 / 36. The populated off-diagonal cell (low coverage, near) shows that coverage and position are not redundant. **(b)** Regressing alignment (JCA) on editorial breadth, log size, and intrinsic shape leaves a substantial share of variance unexplained: R^2 = 0.395 for the entropy specification and R^2 = 0.496 for the concentration specification (N = 521).

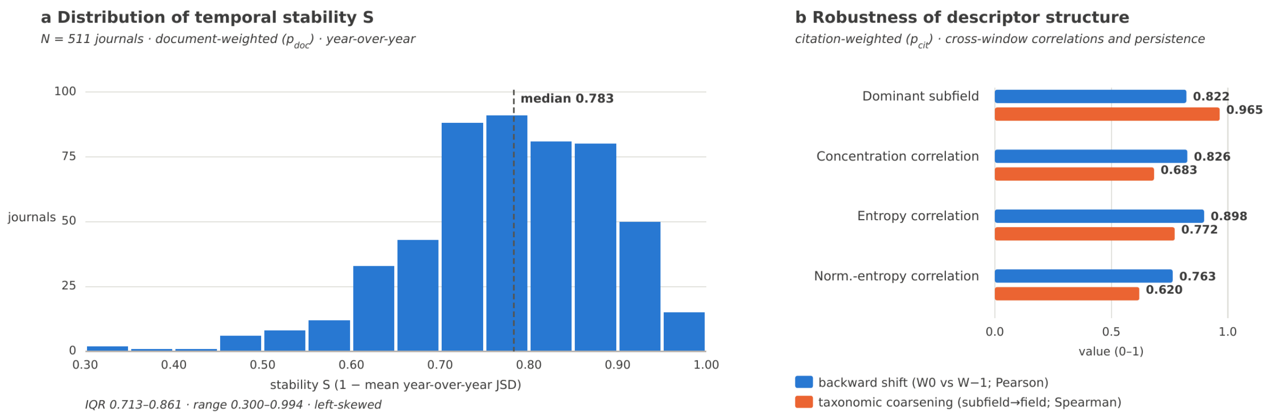

Figure 4 (produced; vector PDF available). *Temporal stability and robustness of descriptor structure.* **(a)** Distribution of year-over-year stability S over the analytic cohort (N = 511, document-weighted); left-skewed with median 0.783 (IQR 0.713–0.861; range 0.300–0.994). **(b)** Under a backward temporal shift (W0 vs W−1; Pearson) and a taxonomic coarsening (subfield → field; Spearman), both citation-weighted, dominant-subfield persistence and the cross-window shape correlations remain high, indicating the descriptor structure is preserved.

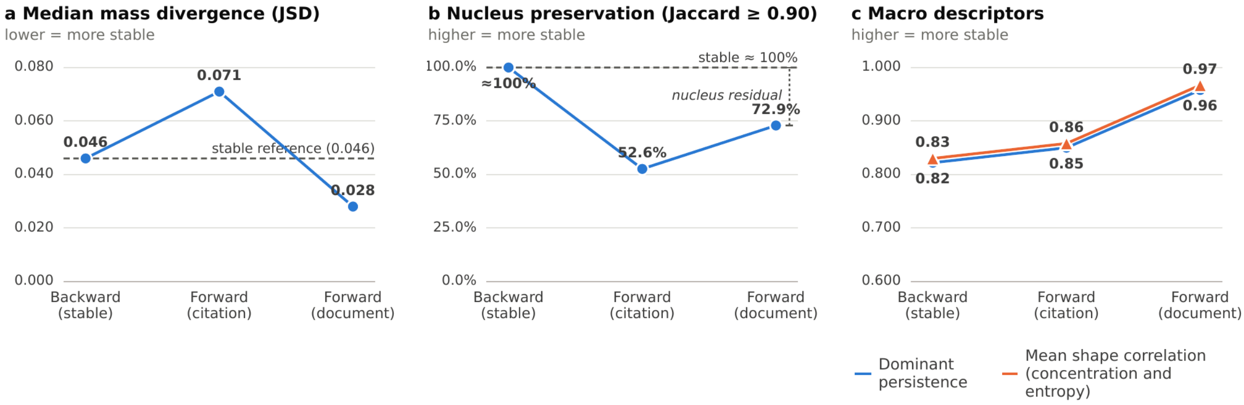

Figure 5 (flagship — produced; vector PDF available). *Backward reference and forward replication under two weighting schemes.* Each panel follows a quantity across the stable backward reference, the forward citation-weighted replication, and the forward document-weighted control. Forward results use the common panel of 506 journals. **(a)** Median Jensen–Shannon divergence increases under citation weighting but falls below the stable backward reference under document weighting. **(b)** Nucleus preservation declines sharply under citation weighting and recovers only partially under document weighting, leaving a nucleus residual relative to the stable reference. **(c)** Dominant-subfield persistence and mean shape correlation remain high, with their strongest values under document weighting. Together, panels (a) and (b) show that the continuous whole-distribution measure returns to the stable range, whereas the threshold-based nucleus measure does not fully recover.

Supplementary Material (provided as a separate document)

Formal statements and proofs of the propositions · exact metric definitions against the canonical implementation (log bases; nucleus tie-breaking) · operationalization of the decision rule and thresholds · preregistration documents (forward replication and documentary control) · code-to-documentation audit report · table of SHA-256 version identifiers.

Back matter

Competing interests. The author is a consultant at RosFlo Limitada and is developing an analytics product (Journal Identity Analytics) based on the method presented here. The preregistered protocol, fixed thresholds, and public reproducibility artifacts (code, data, and a code-to-documentation audit with version hashes) were designed in part to mitigate any associated bias.

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Data and code availability. A reproducibility package supporting this study—comprising all code, data manifests, preregistration documents, and the code-to-documentation audit report—accompanies the manuscript. A Zenodo DOI has been reserved for this package (https://doi.org/10.5281/zenodo.21659764); the public record will become available upon publication of the deposit. Each released artifact—including the canonical implementation, the analysis outputs, the preregistration documents, and the audit report—carries a SHA-256 version identifier, and the version register links every reported quantity to the exact implementation and execution that produced it.

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